

SHASHANK JOSHI

shashank.joshi.c2025@iitbombay.org • +91-9966675787 • GitHub • LinkedIn
AI & Data Professional | GenAI • Analytics • Automation | IIT Bombay

SUMMARY

AI and Data Professional with 2+ years of experience building scalable analytics systems, automated ETL pipelines, and AI-driven workflows across healthcare, political analytics, and computational biology domains. Expert at transforming massive, unstructured datasets into production-ready data and AI solutions.

EDUCATION

Indian Institute of Technology (IIT) Bombay <i>M.Tech in Biomedical Engineering</i>	Jul 2023 – Jul 2025 GPA: 9.21/10
Maharaja Sayajirao University (MSU) Baroda <i>M.Sc in Biotechnology</i>	Nov 2020 – May 2022 GPA: 8.05/10

WORK EXPERIENCE

Indian Political Action Committee (I-PAC) <i>Associate – Data Analytics & Technology</i> - Engineered automated ETL pipelines processing 1M+ daily records, reducing workflow latency from 24 to 4 hours. - Developed automated visual recognition systems using ONNX and Selenium, reducing large-scale manual operational effort. - Built real-time dashboards and monitoring systems supporting analytics across 80K+ operational units. - Designed centralized candidature analysis systems integrating NLP-based sentiment analysis and automated reporting. - Automated cross-functional reporting workflows using Python and Apps Script, saving 5+ manual hours daily.	Nov 2025 – Present Hyderabad / Kolkata
U The Mind Company <i>Molecular Dynamics Expert (Remote)</i> - Built scalable machine learning workflows for biological data forecasting using PyTorch-based pipelines. - Automated multi-step data workflows and scheduling systems using Python and Linux-based orchestration tools.	Apr 2025 – Aug 2025 Ohio, USA
IISER Tirupati <i>Junior Research Fellow</i> - Analyzed 33 large-scale TCGA and ICGC cohorts for biomarker discovery using survival analysis and gene expression profiling. - Executed molecular dynamics simulations and protein stability benchmarks using GROMACS and DynaMut.	Aug 2022 – Jul 2023 Tirupati, India

SKILLS

Programming & Analytics: Python, SQL, Excel, Pandas, Scikit-learn, Power BI, Google Apps Script
AI / ML / GenAI: PyTorch, TensorFlow, ONNX, LangChain, LangGraph, RAG, Prompt Engineering, NLP
Data Engineering & Cloud: ETL/ELT Pipelines, MySQL, AWS (S3, Lambda, Bedrock), Docker, Git
Deployment & Automation: Streamlit, FastAPI, Linux, Selenium, Workflow Automation
Domain Specific: Medical Imaging, Multi-omics, NGS Analysis, Biopython, pydeseq2

KEY PROJECTS

Autonomous GenAI Data Engineering Agent ↗ Built an autonomous agent for schema inference and transformation; processed 100K+ records into analytics-ready Parquet datasets with automated validation workflows.	LangGraph Pydantic
Medical Imaging with Deep Learning ↗ Implemented Attention UNET and Residual UNET architectures for multimodal medical image segmentation, improving segmentation performance and boundaries across healthcare datasets.	TensorFlow PyTorch
NGS Analysis Pipeline for Sickle Cell Anaemia ↗ Developed automated variant analysis workflows and cloud sequence processing infrastructure to optimize data velocity for scalable genomics.	Biopython Pandas

ACHIEVEMENTS & LEADERSHIP

- National Ranks:** AIR 53 in GAT-B 2020 (Top 0.5%) | AIR 75 in IIT JAM 2021 | AIR 216 in GATE 2021.
- Awarded **PG Sportsman of the Year 2024–25** at IIT Bombay for excellence in endurance sports.
- Symbiotek Council (IIT Bombay):** Website Manager & Marketing Coordinator; built responsive web architectures.
- Academic Representative (PGAC, IIT Bombay):** Represented and coordinated academic concerns for 60+ postgraduates.

LANGUAGES & INTERESTS

Languages: English (Fluent), Hindi (Fluent), Marathi, Kannada, Telugu
Interests & Hobbies: Marathons, Cycling, Swimming, Fitness, Playing Guitar, Trekking, Exploring new places